

Simultaneous Input and State Estimation for Integrated Motor-Transmission Systems in a Controller Area Network Environment via an Adaptive Unscented Kalman Filter

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Abstract—As the requirements on powertrain efficiency of electric vehicles (EVs) are increasing, integrated motor-transmission (IMT) powertrain systems for EVs are becoming a promising solution. For the integration of IMT powertrain systems, the system state information and the actuator status are usually required for the closed-loop controller design or the on-board fault diagnosis. Embracing the demands, an observer for simultaneous estimation of input and system state of an IMT powertrain system is studied in this paper. It is well-known that controller area network (CAN) has been dominant in the vehicle network, which is used to communicate among controllers, sensors, and actuators. However, the CAN bus always induces time-varying delays when there are a number of communication nodes on the bus. The CAN-bus induced delay would result in vibrations in the vehicle powertrain or even deterioration of the entire closed-loop system. To deal with the CAN-bus induced delay in the estimation work for IMT powertrain systems, the potential random delays are considered in a three-state nonlinear model which represents the behavior of an IMT system. To estimate the input and state simultaneously, an adaptive unscented Kalman filter (AUKF) is adopted. As we know, the adopted AUKF has the benefits of dealing with system nonlinearities and calculating the noise covariance matrix automatically. Simulations and comparisons are carried out. We can see from the results that the proposed observer estimates the input and system state well. Moreover, the resulting estimation error is smaller comparing

with the estimation error of the observer based on extended Kalman filter algorithm.

Index Terms—Adaptive unscented Kalman filter (AUKF), integrated motor-transmission (IMT), network-induced delays, vehicle applications.

I. INTRODUCTION

ELECTRIC vehicles (EVs) are steadily replacing traditional gasoline vehicles. Due to the irreversible trend, the research on EVs has been extensive in recent years [1]–[4]. As an important component for EVs, integrated motor-transmission (IMT) systems have attracted more attention since the size is smart and the efficiency is high for commercial EVs [5]–[7]. As the driving motor and transmission can be coupled directly by coordination control in an IMT system, the clutch becomes unnecessary. Thus, the whole cost of the system can be reduced thanks to the smaller motor size and simplified powertrain structure [8]. While enjoying the benefit of the IMT system, driveline oscillation may be a problem induced by canceling the clutch [9]. Generally, the rotational speed of the motor, the rotational speed of the wheel, and the output status of actuator are regarded as the most important state information for the closed-loop controller design and the fault diagnosis in an IMT powertrain system [10]. In the literature, a model-based observer was usually utilized to estimate the system state or input (see the papers and references in [11]–[17]). The main principle of observer design is to employ the system output to construct the system state or input. In the traditional case, the system output is directly transmitted to the observer as the input. However, for current vehicular applications, the controller area network (CAN) bus has been widely used to share the messages among different nodes [18], [19]. Though the CAN bus is quite convenient, it may result in a message delay when the number of communication nodes is large and the communication bandwidth is limited. The phenomenon becomes worse if the priority of the communication node is low [20]–[22]. The random communication delays may have a serious impact on the observer system and even have tremendous consequences for the brake system of high speed moving vehicles [23]. Consequently,

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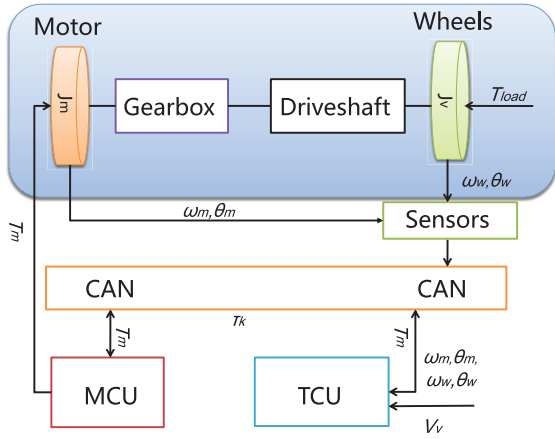


Fig. 1. Schematic of an IMT powertrain system in a CAN environment.

the CAN-bus-induced time-varying random delays should be modeled and considered for the estimator design in the IMT powertrain system.

On the other hand, the simultaneous estimation of system inputs and state has become a very important issue in the area of control and applications [24]–[27]. In [24], the proposed input-and-state estimation method is only applicable to linear discrete-time stochastic systems. For time-varying linear systems, we can see the estimation work in [28] and [29]. In order to deal with nonlinear systems, which are more general than the linear ones, extended Kalman filter (EKF) and unscented Kalman filter (UKF) algorithms were developed gradually [30], [31]. Jiang *et al.* [32] utilized the method of EKF to estimate the system input and system states. We can see from the simulation response that the EKF-based observers work well for the diesel-engine selective catalytic reduction systems. However, in the course of the EKF algorithm, the initial noise covariance matrices should be given. Mismatched initial noise covariance matrices may deteriorate the designed estimator. But the fact is that precise initial noise covariance matrices are too difficult to obtain. Guan and Saif [33] employed an approach to estimate the unknown input states of linear time-invariant systems. Nevertheless, the designed observer can only be suitable for linear systems. A novel method for the design of input observer can be seen in [34] which was aimed at solving the problem for nonlinear systems. However, the proposed method could only be adopted to nonlinear systems with special forms.

In order to deal with both network induced delay and non-linearity in powertrain system well, in this paper we focus on input and state simultaneous observer design for an IMT powertrain system by utilizing an adaptive AUKF algorithm. Since the CAN bus is involved in the powertrain system, the CAN-bus-induced delay is considered during the system output delivery. The proposed observer has the function of estimating the input motor torque and rotational speed of the wheel simultaneously. To avoid the challenges on the selection of initial noise covariance matrices in many algorithms such as EKF and UKF, the adaptive noise statistics estimator is embedded and the AUKF is used in this paper. Thanks to the adaptive property of AUKF, we can simplify the calculation

of noise covariance matrices [35] and improve the estimation performance for IMT powertrain system. The two main contributions of this paper are as follows.

- 1) The CAN-bus-induced delay is considered in the proposed observer, which is very important to improve the estimation accuracy.
- 2) A novel method of adaptive UKF (AUKF) is employed to estimate the input and state information of IMT powertrain system, and simulation results prove that the performance of AUKF is better than that of EKF significantly.

The rest of this paper is organized as follows. In Section II, problem formulation including IMT powertrain and network-induced random delays models are built. In Section III, the AUKF-based observer design approach for IMT systems is proposed. In Section IV, simulation and comparison results carried out in MATLAB/Simulink are given to show the performance of the designed AUKF-based observer. Section V is the conclusion section.

II. PROBLEM FORMULATION

The CAN-based structure of an IMT powertrain system is described as Fig. 1. The powertrain mainly includes motor, gearbox, driveshaft, and wheels. The powertrain state information measured by sensors is sent through CAN to transmission control unit (TCU) periodically. Then, the TCU could determine required motor torque according to the messages on the CAN bus and current vehicle speed. The required motor torque determined by the TCU is transmitted on the CAN bus as well. When the motor control unit (MCU) receives the required torque, the motor is actuated to achieve the target motor torque [36]. It should be noted that the main uncertain disturbance comes from the external load torque T_{load} and the CAN-bus-induced delay τ_k . For the modeling accuracy, both terms should be included in modeling of the IMT powertrain system.

A. IMT Powertrain Operation Principle

According to the properties of vehicle dynamics and motor-gearbox set, the mathematical model of IMT powertrain system is established [37]. The differential equations are described as follows:

$$J_m \dot{\omega}_m = T_m - T_f / i_g i_0 - c_m \omega_m \quad (1)$$

$$J_{mg} = J_m + J_g / i_g^2 i_0^2 \quad (2)$$

$$J_v \dot{\omega}_w = T_f - T_{lo} \quad (3)$$

$$T_f = c_f (\omega_g - \omega_w) + k_f (\theta_g - \theta_w) \quad (4)$$

$$T_{lo} = T_{ro} + T_{gd} + T_{ad} \quad (5)$$

$$\omega_m = i_g i_0 \omega_g \quad (6)$$

$$\theta_m = i_g i_0 \theta_g \quad (7)$$

in which T_m represents the motor torque; T_f refers to the torque of the driveshaft; T_{lo} expresses the external load torque which includes resistant torque T_{gd} resulted from road grade, roll torque T_{ro} , and air drag T_{ad} ; J_m is motor inertia and J_{mg} is the equivalent inertia of the motor; J_g denotes the gearbox inertia; J_v indicates the vehicle inertia; ω_m , ω_w , and ω_g depict the

rotational speed of motor, wheel, and gearbox, respectively, meanwhile, θ_m , θ_w , and θ_g are the output angles; c_m stands for the motor's damping coefficient; c_f is named as driveshaft damping coefficient; and k_f is described as a stiffness factor.

Since the value of air drag is small in the system, the non-linear variable can be approximated as a linear equation [36], which is written as follows:

$$T_{ad} = c_a \omega_w \quad (8)$$

where c_a denotes an approximation factor. The linearization error is added into the process noise, which could be eliminated by Kalman filter in this paper.

The output angles and rotational speed of motor with wheel are selected as the state variables. In order to facilitate expression, the state variable of output angles is defined as θ_{mw} . Then, with consideration of all the dynamics, a three-state model has the following expression:

$$\begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} &= \begin{bmatrix} \dot{\omega}_m \\ \dot{\omega}_w \\ \dot{\theta}_{mw} \end{bmatrix} \\ &= \begin{bmatrix} -\left(\frac{c_m}{J_{mg}} + \frac{c_f}{i_g^2 i_0 J_{mg}}\right) \omega_m + \frac{c_f}{i_g i_0 J_{mg}} \omega_w \\ \frac{c_f}{i_g i_0 J_v} \omega_m - \frac{c_a + c_f}{J_v} \omega_w + \frac{k_f}{J_v} \left(\frac{\theta_m}{i_g i_0} - \theta_w\right) - \frac{T_{ro} + T_{gd}}{J_v} \\ \frac{1}{i_g i_0} \omega_m - \omega_w \\ -\frac{k_f}{i_g i_0 J_{mg}} \left(\frac{\theta_m}{i_g i_0} - \theta_w\right) \\ +0 \\ +0 \end{bmatrix} + \begin{bmatrix} \frac{1}{J_{mg}} \\ 0 \\ 0 \end{bmatrix} T_m + d \end{bmatrix} \quad (9)$$

where $\theta_{mw} = \theta_m / i_g i_0 - \theta_w$; d depicts the external disturbance including the model errors in load torque and the linearization error [9]. Besides, T_m is regarded as the input of the system.

B. Modeling of Network-Induced Random Delays

In this paper, the CAN-bus-induced delays are assumed to be bounded and the upper bound is known *a priori*. Then, we can illustrate the signals and delays among different communication nodes shown in Fig. 2. First, the measurements are generated from sensors with a fixed sampling period T . Second, the measurements are sent to the TCU node through CAN, and the signal arrived at t_1 . Then the TCU would calculate the input motor torque u_k as soon as possible based on the measurements. Third, the command signal of then motor torque would be sent to MCU node via CAN and arrived at t_2 . Meanwhile, the powertrain system starts to work and supply the required torque. Therefore, according to the analysis above, the delay of step k can be described as τ_k in Fig. 2. The control input of the powertrain system with delay can be written as follows [36]:

$$u(t) = u_k \quad \forall t \in [t_k + \tau_k, t_{k+1} + \tau_{k+1}] \quad (10)$$

in which $t_k = kT$ expresses the beginning of each step.

A reasonable assumption is that the CAN-induced delay is bounded. Then, we can define the maximum delay in the following form:

$$\tau_{\max} = (\gamma + \nu)T \quad (11)$$

where $\gamma \in \mathbb{Z}_+$, and $\nu \in R_{[0,1)}$. Then, the discrete-time model of the IMT powertrain system with the CAN-bus-induced

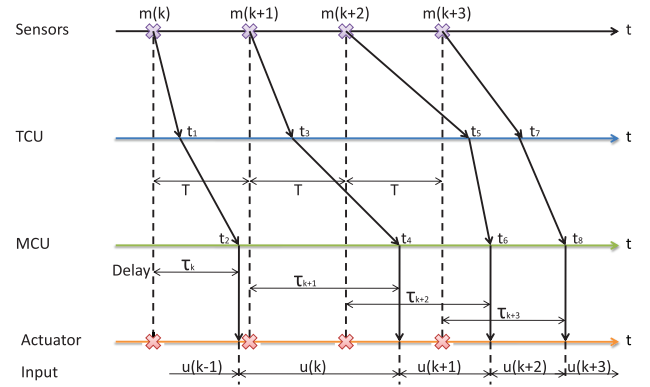


Fig. 2. Schematic of the CAN-induced time-varying delays [9].

delay is represented as follows [9]:

$$\begin{aligned} x(k+1) &= \bar{A}x(k) + \bar{B}u_k + \bar{D}d_k + \Delta_{0,k}(u_{k-1} - u_k) \\ &\quad + \Delta_{1,k}(u_{k-2} - u_{k-1}) + \dots \\ &\quad + \Delta_{\gamma,k}(u_{k-\gamma-1} - u_{k-\gamma}) \end{aligned} \quad (12)$$

where

$$\bar{A} = e^{\bar{A}T}, \quad \bar{B} = \int_0^T e^{\bar{A}(T-\theta)} d\theta \bar{B}, \quad \bar{D} = \int_0^T e^{\bar{A}(T-\theta)} d\theta \bar{D} \quad (13)$$

$$\Delta_{i,k} = \begin{cases} 0 & \tau_{k-i} - iT \leq 0 \\ \int_0^{\tau_{k-i}-iT} e^{\bar{A}(T-\theta)} d\theta \bar{B} & 0 < \tau_{k-i} - iT < T \\ \int_0^T e^{\bar{A}(T-\theta)} d\theta \bar{B} & T \leq \tau_{k-i} - iT. \end{cases} \quad (14)$$

According to the communication protocol of CAN [19] and the CAN analysis work in [20] and [21], there are two properties for the CAN-bus-induced delay as

$$\tau_k \geq \tau_{k-1} - T \quad (15)$$

and

$$\tau_{\max,j} \leq \frac{(j+2)l}{R - \sum_{i=0}^{j-1} \frac{1}{c_i}} \quad (16)$$

in which $\tau_{\max,j}$ stands for the maximum delay of j th priority among CAN messages; R expresses the CAN transmission rate; l is named as the maximal frame length; and c_i denotes the cycle length for the i th priority message.

To simplify the system analysis and observer design, linearization of nonlinearities is necessary in many applications. Adopting the works in [9] and [38], Taylor expansion is employed to explore the nonlinearities in the equations. The formulation of expansion is described as follows:

$$\begin{aligned} \Delta(x) &= \int_0^T e^{\bar{A}(T-\theta)} d\theta \\ &= \Delta(0) + \dot{\Delta}(0)x + \ddot{\Delta}(0)\frac{x^2}{2} + \dots + \frac{d^n \Delta(0)x^n}{dx^n} \frac{x^n}{n!} \\ &\quad + \sum_{n=1}^h \frac{-1^{n+1}}{n!} \bar{A}^{n-1} e^{\bar{A}T} x^n + h.o.t \end{aligned} \quad (17)$$

where h denotes the selected order. *h.o.t* represents the higher-order term. It should be pointed out that, in a Taylor expansion, the higher-order term in (17) can be ignored when the value for it is small enough.

While $\Delta(0) = 0$, the (17) can be simplified as

$$\begin{aligned} \Delta(x) &= A_\tau x_\tau \\ A_\tau &= e^{AT} \begin{bmatrix} \frac{(-1)^2}{1!} A^0 & \frac{(-1)^3}{2!} A^1 & \dots & \frac{(-1)^{h+1}}{h!} A^{h-1} \end{bmatrix} \\ x_\tau &= [x \quad x^2 \quad \dots \quad x^h]^T. \end{aligned} \quad (18)$$

As the variables are restricted in $[0, T]$ or $[0, \nu T]$ with any delay τ_k in the nonlinear system, the vertices of polytope for the delays can be defined as follows:

$$\begin{aligned} x_{\tau,j,0} &= [\tilde{x}_\tau \quad \tilde{x}_\tau^2 \quad \dots \quad \tilde{x}_\tau^h]^T \\ x_{\tau,j,1} &= [\tilde{x}_{\tau,j} \quad \tilde{x}_{\tau,j}^2 \quad \dots \quad \tilde{x}_{\tau,j}^h]^T \\ &\vdots \\ x_{\tau,j,h} &= [\tilde{x}_{\tau,j} \quad \tilde{x}_{\tau,j}^2 \quad \dots \quad \tilde{x}_{\tau,j}^h]^T \\ j &= 0, 1; \quad \tilde{x}_\tau = 0; \quad \tilde{x}_{\tau,0} = T; \quad \tilde{x}_{\tau,1} = \nu T. \end{aligned} \quad (19)$$

Therefore, the polytope vertices in the matrices $\Delta_{i,k}$ could be described as follows:

$$\begin{aligned} \hat{\Delta}_{0,n} &= A_\tau x_{\tau,0,n} B \\ \hat{\Delta}_{1,n} &= A_\tau x_{\tau,1,n} B \\ \forall n &= 0, 1, \dots, h. \end{aligned} \quad (20)$$

In addition, the uncertain matrices in (12) with time-varying delays can be refined in the following form:

$$\begin{aligned} \Delta_{i,k} &= \sum_{n=0}^h \xi_{i,n}(k) \hat{\Delta}_{1,n} B \quad \forall i = 0, 1, \dots, \gamma - 1 \\ \Delta_{i,k} &= \sum_{n=0}^h \xi_{i,n}(k) \hat{\Delta}_{0,n} B, i = \gamma \\ \sum_{n=0}^h \xi_{i,n}(k) &= 1 \end{aligned} \quad (21)$$

in which $\xi_{i,n}(k)$ stands for a time-varying coefficient which is relevant to delay τ_k .

Remark 1: In the literature, there are several approaches to deal with sampled-data systems [39], [40]. It infers from [39] that the matrix exponential term in (17) can be transformed into a tractable matrix with the help of confluent Vandermonde matrix. Different from the work in [39] and [40], in this paper, we employ the Taylor expansion and ignore the high-order term. The main reason is that, for electrical and mechanical systems, we should tradeoff between the computational load and the performance. Based on the experience on several mechatronic systems, the performance of this approximation approach is acceptable.

III. ESTIMATION OF INPUT MOTOR TORQUE AND STATE

Initially, Kalman filter is well known for the orbit prediction, which is found by Kalman and some other researchers in 1960s [28]. To deal with nonlinear system, EKF and UKF are proposed. Actually, EKF employs Taylor expansion to linearize the nonlinear model. However, EKF is overwhelmingly dependent on the mathematical model of systems. When the

model is subject to uncertainties, the performance of the filter cannot be guaranteed. What is more, the Jacobian matrices in EKF algorithm is too complex to calculate, which would increase the computational burden [41]. In this paper, both the model errors due to load torque and the linearization error of air drag would degrade the precision and reliability of system. Thus, the EKF algorithm is not suitable to estimate the actuator fault and states.

To overcome the shortcomings of EKF, the UKF is developed from unscented transformation and Kalman filter. UKF could avoid the complex calculation of Jacobian matrices [42]. Moreover, compared with EKF, the degree of dependence on mathematical model is lower in UKF algorithm.

We can see from the EKF and UKF algorithms that the noise covariance matrices are very important for the algorithms. However, the initial noise covariance matrices are also difficult to obtain in many practical applications [43]. Moreover, in many cases, the process and measurement noises are unknown or even time-varying. Generally, it is time consuming to search covariance matrices for the process and measurement noises in EKF and UKF algorithms. Nevertheless, unsuitable noise covariance matrices would also deteriorate the performance of KF-based observers [44]. Therefore, adaptive control method is developed in Kalman filter to address the problem. The adaptive algorithm could estimate the noise covariance matrices according to the filter learning history, and this algorithm could also calculate the noise covariance matrices online during working process [42].

A. UKF Algorithm

The UKF algorithm and calculation procedure are described as follows.

1) Predicting Step:

a) *Initialization:* Suppose that we have the following discrete-time state-space model:

$$x(k) = f[x(k-1), u(k)] + w(k) \quad (22)$$

$$z(k) = h[x(k)] + v(k) \quad (23)$$

where $f(x)$ refers to the predicted function, $x(k)$ denotes the state vector, $u(k)$ stands for the input vector, and $w(k)$ is named as the Gaussian process noise whose covariance is $Q(k)$. In (23), $h(x)$ represents the measurement function, and $v(k)$ states the Gaussian measurement noise whose covariance is $R(k)$. It should be pointed out that $w(k)$ and $v(k)$ are independent Gaussian white noises.

At the first step, the initial state vector of the nonlinear system is selected as the mean value of x_0 and the error covariance is represented as P_0 . The equations are shown as follows:

$$\hat{x}_0 = E[x_0] \quad (24)$$

$$P_0 = E[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T]. \quad (25)$$

b) *Calculating sigma points:* On the basis of $\hat{x}_{k-1}$ and the error covariance P_{k-1} , a series of weighted samples which are named as sigma points are calculated according to the

following equation:

$$X_{k-1}^{(i)} = \begin{cases} \hat{x}_{k-1}, & i = 0 \\ \hat{x}_{k-1} + (\sqrt{(n_x + \lambda)P_{k-1}})_{i-n_x}, & i = 1, \dots, n_x \\ \hat{x}_{k-1} - (\sqrt{(n_x + \lambda)P_{k-1}})_{i-n_x}, & i = n_x + 1, \dots, 2n_x \end{cases} \quad (26)$$

where n_x stands for the state vector dimension, λ denotes a scaling factor, and its expression can be described as follows:

$$\lambda = \alpha^2(n_x + q) - n_x. \quad (27)$$

Here, α depicts a positive constant, q is called as the secondary scaling factor set to 0 generally. Then the mean weights W_m can be computed by using the following expression:

$$W_m^{(i)} = \begin{cases} \frac{\lambda}{\lambda + n_x}, & i = 0 \\ \frac{\lambda}{2(\lambda + n_x)}, & i = 1, \dots, 2n_x. \end{cases} \quad (28)$$

c) *Propagating sigma points*: At the third step, these sigma points are propagated based on the predicted function $f(x)$

$$X_{k|k-1}^{(i)} = f(X_{k-1}^{(i)}, u_{k-1}). \quad (29)$$

d) *A Priori state and error covariance calculation*: We can estimate *a priori* state as

$$\hat{x}_{k|k-1} = \sum_{i=0}^{2n_x} W_m^{(i)} X_{k|k-1}^{(i)}. \quad (30)$$

A priori error covariance is based on the variance weights W_c which is expressed with the following equation:

$$W_c^{(i)} = \begin{cases} \frac{\lambda}{\lambda + n_x} + (1 - \alpha^2 + \beta), & i = 0 \\ \frac{\lambda}{2(\lambda + n_x)}, & i = 1, \dots, 2n_x \end{cases} \quad (31)$$

where β stands for a parameter in state estimation. In this paper, β is set to 2 for getting the optimal estimation under Gaussian distributions. Then, *a priori* error covariance can be calculated as the following equation:

$$P_{k|k-1} = \sum_{i=0}^{2n_x} W_c^{(i)} (X_{k|k-1}^{(i)} - \hat{x}_{k|k-1}) (X_{k|k-1}^{(i)} - \hat{x}_{k|k-1})^T + Q_k. \quad (32)$$

2) Updating Process:

a) *Calculating sigma points*: During the predicting process, there are a group of sigma points as well. We should update the sigma points by *a priori* state estimated equation with the following expression:

$$X_{k|k-1}^{(i)} = \begin{cases} \hat{x}_{k|k-1}, & i = 0 \\ \hat{x}_{k|k-1} + (\sqrt{(n_x + \lambda)P_{k|k-1}})_{i-n_x}, & i = 1, \dots, n_x \\ \hat{x}_{k|k-1} - (\sqrt{(n_x + \lambda)P_{k|k-1}})_{i-n_x}, & i = n_x + 1, \dots, 2n_x. \end{cases} \quad (33)$$

b) *System output prediction*: When computing the new sigma point by using the measurement function $h(x)$, we can get the sigma point output with the following formula:

$$Y_{k|k-1}^{(i)} = h(X_{k|k-1}^{(i)}, u_k). \quad (34)$$

With the expression in (34), the predicted system output is expressed as

$$\hat{Y}_{k|k-1} = \sum_{i=0}^{2n_x} W_m^{(i)} Y_{k|k-1}^{(i)}. \quad (35)$$

c) *Determining the value of Kalman gain K_k* : At this step, the main work is to calculate the Kalman gain K_k on the basis of predicted output covariance matrix $P_{y_k y_k}$ and cross-covariance $P_{x_k y_k}$. The calculations are represented in the following equations:

$$P_{y_k y_k} = \sum_{i=0}^{2n_x} W_c^{(i)} (Y_{k|k-1}^{(i)} - \hat{Y}_{k|k-1}) (Y_{k|k-1}^{(i)} - \hat{Y}_{k|k-1})^T + R_k \quad (36)$$

$$P_{x_k y_k} = \sum_{i=0}^{2n_x} W_c^{(i)} (X_{k|k-1}^{(i)} - \hat{x}_{k|k-1}) (Y_{k|k-1}^{(i)} - \hat{Y}_{k|k-1})^T \quad (37)$$

$$K_k = P_{x_k y_k} P_{y_k y_k}^{-1}. \quad (38)$$

d) *Computing a posteriori state and error covariance*: With the assists of the measurement information and determined Kalman gain, the corresponding *a posteriori* state and error covariance estimation are computed as

$$\hat{x}_k = \hat{x}_{k|k-1} + K_k (Z_k - \hat{Y}_{k|k-1}) \quad (39)$$

$$P_k = P_{k|k-1} - K_k P_{y_k y_k} K_k^T \quad (40)$$

where Z_k stands for the sensor reading at step k . In order to run the iterative algorithm for the next step, *a posteriori* error covariance is also essential.

B. Adaptive Algorithm

The adaptive algorithm can be depicted as follows [35]:

$$Q_k = Q_{k-1} + d_{k-1} (K_k \varepsilon_k \varepsilon_k^T K_k^T + P_k - P_{k|k-1}) \quad (41)$$

$$R_k = R_{k-1} + d_{k-1} (\varepsilon_k \varepsilon_k^T - P_{y_k y_k}) \quad (42)$$

$$d_k = (1 - b) / (1 - b^{k+1}) \quad (43)$$

$$\varepsilon_k = Z_k - h(\hat{x}_k) \quad (44)$$

in which d_k denotes a scaling parameter, b is named as a forgetting factor and usually it is set between 0.95 and 0.995, and ε_k represents the error between the sensor measurement and the estimation at the k th step.

C. Estimation of Input Motor Torque and State

Though the input motor torque is varying at different operation points, we can assume that the value of the torque does not change too frequently. Consequently, the derivative of the input motor torque can be assumed to be zero as

$$\dot{T}_m = 0. \quad (45)$$

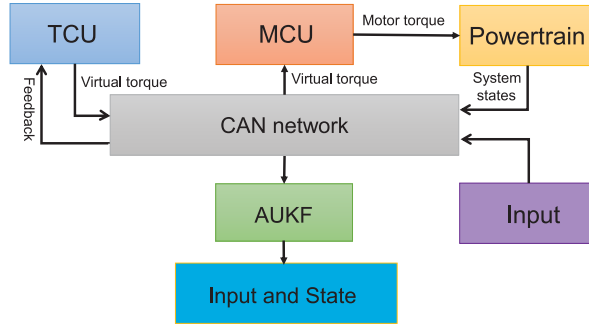


Fig. 3. Simulation of the AUKF-based observer for the IMT powertrain system.

In this paper, we aim to estimate the input motor torque and rotational speed of the wheel. Then the prediction state space equations can be described as follows:

$$\begin{aligned}\hat{x}(k|k-1) &= \begin{bmatrix} \hat{\omega}_w(k|k-1) \\ \hat{T}_m(k|k-1) \end{bmatrix} = f[\hat{x}(k-1|k-1)] \\ &= \begin{bmatrix} \hat{\omega}_w(k-1|k-1) + \Delta t \dot{\hat{\omega}}_w(k-1|k-1) \\ \hat{T}_m(k-1|k-1) \end{bmatrix}\end{aligned}\quad (46)$$

where Δt represents the updating time in the AUKF algorithm. The rotational speed of motor ω_m from sensor reading and calculated output angles θ_{mw} are selected as the measurement equations. Since the CAN-bus-induced delay is mainly dependent on the message priority, it is assumed that the sensor readings have relatively high priority and there is no comparable delay for the readings. Then, the measurement equations are expressed as

$$\begin{aligned}z(k) &= \begin{bmatrix} \omega_m(k) \\ \theta_{mw}(k) \end{bmatrix} \\ &= \begin{bmatrix} \omega_m(k-1) + \Delta t \dot{\omega}_m(k-1) \\ \theta_{mw}(k-1) + \Delta t \dot{\theta}_{mw}(k-1) \end{bmatrix}\end{aligned}\quad (47)$$

where

$$\begin{aligned}\dot{\omega}_m(k) &= -\left(\frac{c_m}{J_{mg}} + \frac{c_f}{i_g^2 i_0 J_{mg}}\right) \omega_m + \frac{c_f}{i_g i_0 J_{mg}} \hat{\omega}_w(k|k-1) \\ &\quad - \frac{k_f}{i_g i_0 J_{mg}} \left(\frac{\theta_m}{i_g i_0} - \theta_w\right) + \hat{T}_m(k|k-1)/J_{mg}\end{aligned}\quad (48)$$

$$\dot{\theta}_{mw}(k) = \frac{1}{i_g i_0} \omega_m - \omega_w(k|k-1).\quad (49)$$

In terms of the experience on the application, we select the initial noise covariance matrices as

$$P_0 = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}, Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, R = \begin{bmatrix} 0.01 & 0 \\ 0 & 1 \end{bmatrix}.\quad (50)$$

Besides, the initial motor torque and rotational speed of the wheel are all set as zero. Then, based on the prediction and measurement equations, the input motor torque and the rotational speed of the wheel can be estimated by using the mentioned AUKF algorithm.

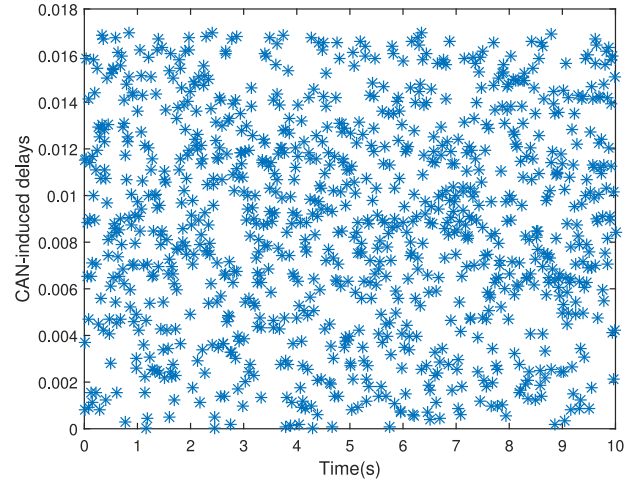


Fig. 4. CAN-bus-induced time-varying delays in the simulations.

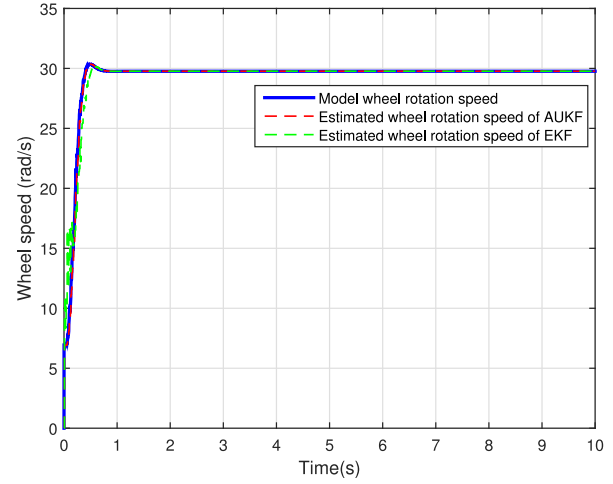


Fig. 5. Comparison of estimation for wheel rotational speed between EKF and AUKF under a step speed.

IV. SIMULATION RESULTS

In this section, simulations and comparisons of the proposed AUKF-based observer for input and state simultaneous estimation are offered with the assistance of MATLAB/Simulink. The schematic structure of simulation model is depicted in Fig. 3. Most of signals are transmitted on the CAN network. An EKF-based observer is employed to compare the estimation performance. Two different maneuvers of wheel speed are investigated in this simulation.

The sampling period of the IMT powertrain system is selected as $T = 10$ ms which is a typical value in vehicular systems. It is assumed that the maximum CAN-bus-induced delay $\tau_{\max}$ is selected as $1.7T$. Since the delays are time-varying and stochastic, different patterns of delays would affect the simulation results. During the simulations and comparisons, the distribution of delays is kept the same, which is depicted in Fig. 4. We can observe that the delays are stochastic and bounded in the range of $[0, 0.017]$ s.

In the simulation model, the initial value of input torque and rotational speed of the wheel are set as zeros. Two different observers are implemented in the simulation environment. Fig. 5 illustrates the comparison of estimated wheel

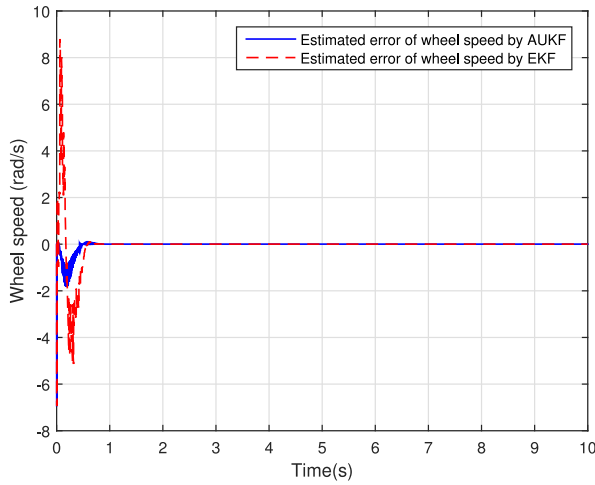


Fig. 6. Estimated errors for wheel rotational speed of AUKF and EKF.

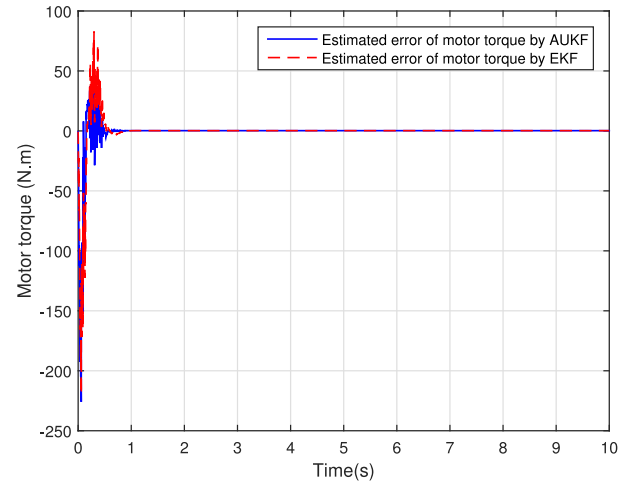


Fig. 8. Estimated errors for input motor torque by using AUKF and EKF.

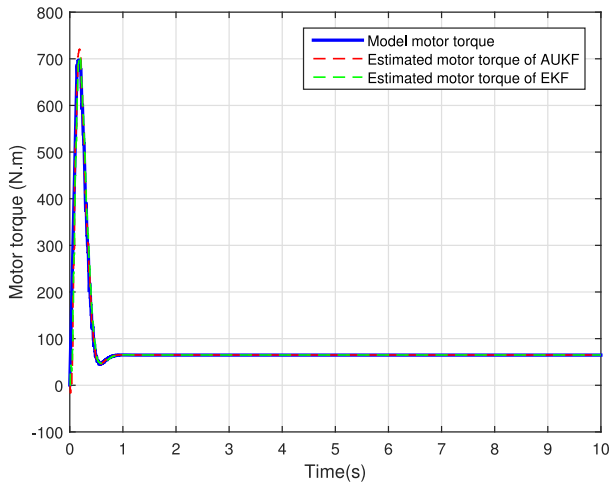


Fig. 7. Comparison of estimation for input motor torque by using EKF and AUKF.

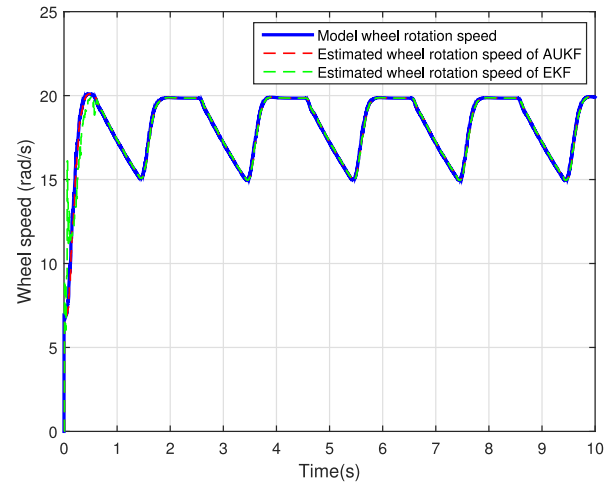


Fig. 9. Comparison of estimation for wheel rotational speed by using EKF and AUKF under a time-varying speed.

rotational speed between EKF-based observer and AUKF-based observer with a step speed. It can be observed that, during the transient response, the AUKF has a better performance on tracking the model value than the EKF-based method. Moreover, the response of the AUKF-based observer is faster. In order to show more direct-viewing comparison of the EKF-based observer and the AUKF-based observer, the errors of the estimated rotational speed are depicted in Fig. 6. In the figure, the smaller estimated error of AUKF prove that the AUKF-based observer could achieve better performance on the observer design work for an IMT powertrain system.

Fig. 7 describes the comparison of estimated input torque of the EKF-based observer and the AUKF-based observer correspondingly. Although the proposed observer cannot match the model input torque during the first 0.2 s due to the difference between initial state and actual value, the estimation tracks the model value rapidly after 0.2 s. What is more, the performance of the AUKF-based observer is better than that of the EKF-based observer in terms of the peak value of the curve. The estimation errors in Fig. 8 also obviously proves that the AUKF-based observer is more accurate than the EKF-based observer.

To further validate the performance of the proposed observer, we have done another simulation in which the wheel speed is not constant but time-varying. The estimations of wheel rotational speed and input motor torque under different conditions are presented. Fig. 9 depicts the estimation of wheel rotational speed under a time-varying condition. Although the speed changes rapidly in this condition, the AUKF-based observer could follow the model speed well. On the contrary, the EKF-based observer could not track the model speed quickly at the beginning of the simulation. Fig. 10 presents the estimation error of the AUKF-observer and the EKF-based error, respectively.

Fig. 11 shows the corresponding estimation comparison of input motor torque between the EKF-based observer and the AUKF-based observer under the time-varying condition. It could be seen that the AUKF-based observer shows better performance especially when the torque varies rapidly, while the EKF-based observer cannot track the model motor torque well at the peak of the curve. Fig. 12 illustrates the estimated errors of the input motor torque in this case. Although the errors of the proposed observers may reach 50 N.m, it also within an

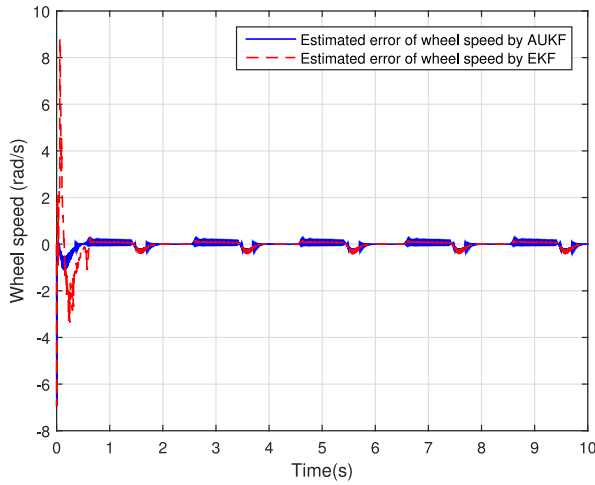


Fig. 10. Estimated errors for wheel rotational speed by using AUKF and EKF.

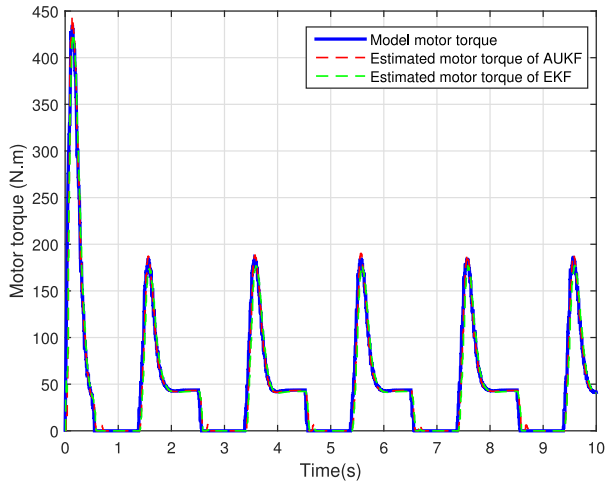


Fig. 11. Comparison of estimation for input motor torque by using EKF and AUKF under a time-varying speed.

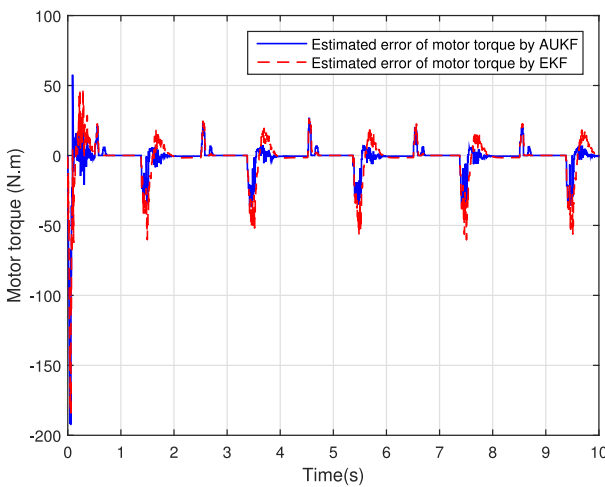


Fig. 12. Estimated errors for input motor torque by using AUKF and EKF.

acceptable range compared with the value of motor torque. Besides, the estimation error of the AUKF-based observer is smaller than that of the EKF-based observer obviously.

TABLE I
RMS OF UKF AND AUKF ESTIMATED ERRORS UNDER A STEP SPEED

Estimation	EKF	AUKF	Relative Improvement
Rotational speed	0.7739	0.2743	64.67%
Motor torque	15.6558	14.0346	10.36%

TABLE II
RMS OF UKF AND AUKF ESTIMATED ERRORS
UNDER A TIME-VARYING SPEED

Estimation	EKF	AUKF	Relative Improvement
Rotational speed	0.5588	0.2526	54.80%
Motor torque	16.2756	13.4122	17.59%

In order to further organize the improvements, two comparison tables are given. Table I denotes the root mean square (RMS) value of errors between the model value and the estimated value of wheel rotational speed and motor torque when the wheel speed is uniform. Table II demonstrates the RMS value of errors between the model value and the estimated value of wheel rotational speed and motor torque under a time-varying speed. As represented in the tables, the RMS value of the EKF errors is always larger than the one of the AUKF errors. The relative improvements are given as well. For the estimation of rotational speed, the relative improvement is more than 50% in both cases. And for the estimation of motor torque, the relative improvement is more than 10% for both simulation cases. Considering the performance of the input torque estimation and the speed estimation, we can claim that the proposed observer works well in the different simulation cases.

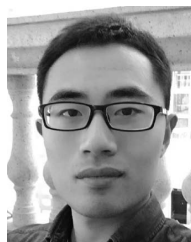
V. CONCLUSION

In this paper, an AUKF-based observer is proposed to estimate input and state simultaneously for IMT powertrain systems which are subject to CAN-bus-induced time-varying delays. It is well-known that the output torque from actuator and wheel rotational speed are very important to fault diagnosis of IMT powertrain systems and the closed-loop controller design. When the CAN-bus-induced delays are involved in the message transmission, the delays are considered in the IMT system models. Then, the AUKF algorithm is employed to obtain the estimation for input motor torque and rotational speed of the wheel. Comparison results between EKF and AUKF show that the AUKF-based estimator can achieve higher estimation accuracy and smaller estimation error. We can conclude that the proposed AUKF-based observer is robust to the nonlinear IMT powertrain system which is under CAN-bus-induced time-varying delays. In the future work, we will focus on the implementation and the observer-based controller design.

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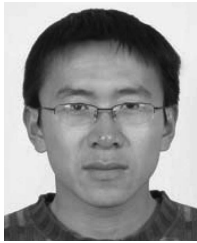
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